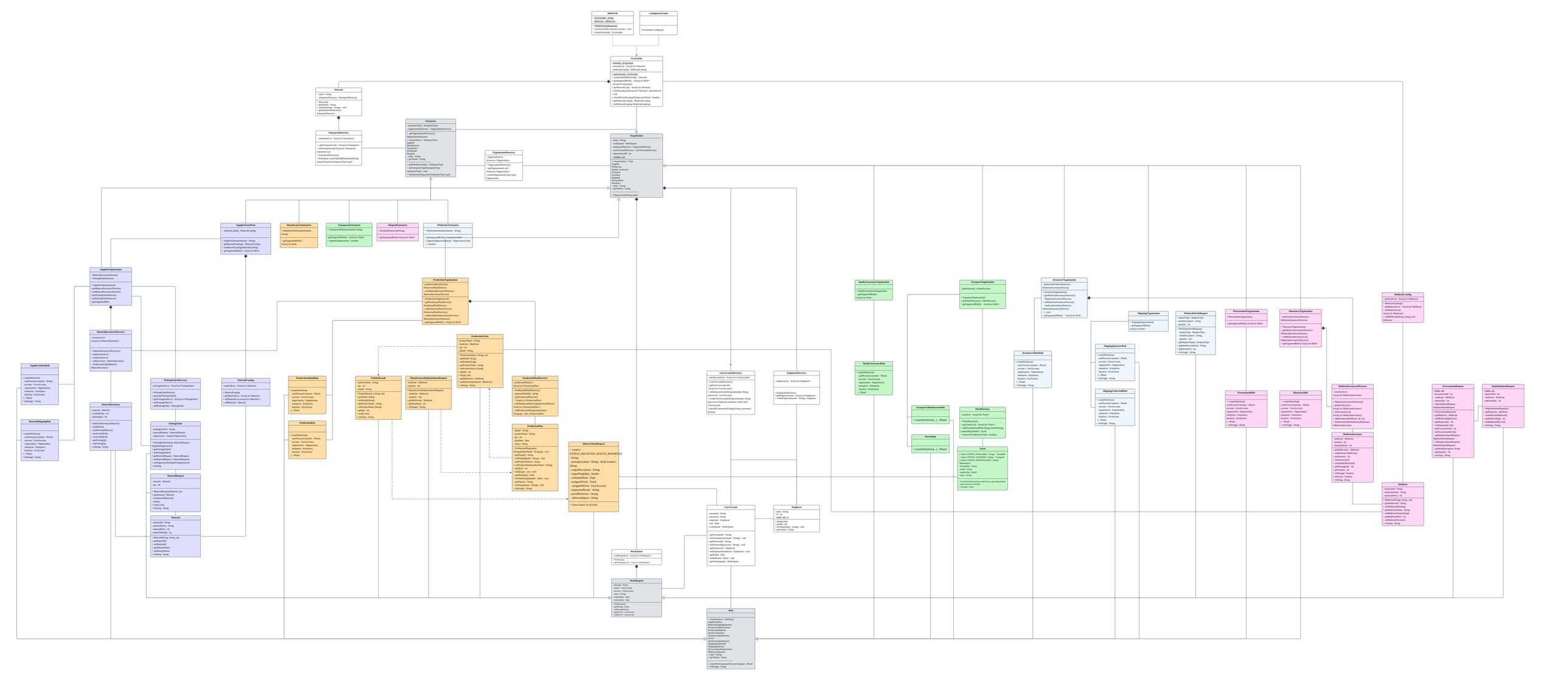

DB4OUtil
- FILENAME : String
- dB4OUtil : DB4OUtil :
+ DB4OUtil getInstance()
+ storeSystem(EcoSystem system) : void
+ retrieveSystem() : EcoSystem
ConfigureASystem
+EcoSystem configure()
EcoSystem
- business : EcoSystem
- networkList : ArrayList<Network>
- medicineCatalog : MedicineCatalog
+ getInstance() : EcoSystem
+ createAndAddNetwork() : Network
+ getSupportedRole() : ArrayList<Role>
- private EcoSystem()
+ getNetworkList() : ArrayList<Network>
+ setNetworkList(ArrayList<Network> networkList) : void
+ checkIfUserIsUnique(String userName) : boolean
+ getMedicineCatalog : MedicineCatalog
+ setMedicineCatalog (MedicineCatalog)
Network
- name : String
- enterpriseDirectory : EnterpriseDirectory
+ Network()
+ getName() : String
+ setName(name : String) : void
+ getEnterpriseDirectory() : EnterpriseDirectory
EnterpriseDirectory
- enterpriseList : ArrayList<Enterprise>
+ getEnterpriseList() : ArrayList<Enterprise>
+ setEnterpriseList(ArrayList<Enterprise> enterpriseList)
+ EnterpriseDirectory()
+ Enterprise createAndAddEnterprise(String name,Enterprise.EnterpriseType type)
Enterprise
- enterpriseType : EnterpriseType
- organizationDirectory : OrganizationDirectory
+ getOrganizationDirectory() : OrganizationDirectory
<<enumeration>> EnterpriseType
Supplier
Manufacturer
Transporter
Wholesaler
Hospital
- value : String
+ getValue() : String

+ getEnterpriseType() : EnterpriseType
+ setEnterpriseType(EnterpriseType enterpriseType) : void
+ Enterprise(String name,EnterpriseType type)
OrganizationDirectory
- organizationList : ArrayList<Organization>
+ OrganizationDirectory()
+ getOrganizationList() : ArrayList<Organization>
+createOrganization(Type type) : Organization
Organization
- name : String
- workQueue : WorkQueue
- employeeDirectory : EmployeeDirectory
- userAccountDirectory : UserAccountDirectory
- organizationID : int
- counter : int
<<enumeration>> Type
Supplier
Production
Quality Assurance
Transport
Inventory
Shipping
Procurement
Pharmacy
- value : String
+ getValue() : String

+ Organization(String name)
SupplierEnterPrise
- materialCatalog : MaterialCatalog
+ SupplierEnterprise(name : String)
+ getMaterialCatalog() : MaterialCatalog
+ setMaterialCatalog(MaterialCatalog)
+ getSupportedRole() : ArrayList<Role>
ManufacturerEnterprise
+ ManufacturerEnterprise(name : String)
+ getSupportedRole() : ArrayList<Role>
TransporterEnterprise
+ TransporterEnterprise(name:String)
+getSupportedRole() : ArrayList<Role>
+ supportsOrganization : boolean
HospitalEnterprise
+ HospitalEnterprise(String)
+ getSupportedRole():ArrayList<Role>
WholesalerEnterprise
+ WholesalerEnterprise(name : String)
+ getSupportedRole() : ArrayList<Role>
+ supportsOrganization(type : Organization.Type) : boolean
SupplierOrganization
- MaterialInventoryDirectory
- PickingOrderDirectory
+ SupplierOrganizaztion()
+getMaterialInventoryDirectory
+setMaterialInventoryDirectory
+getPickingOrderDirectory
+setPickingOrderDirectory
+getSupportedRole
MaterialInventoryDirectory
- inventoryList ArrayList<MaterialInventory>
+ MaterialInventoryDirectory()
+ getInventoryList
+ setInventoryList
+ addInventory : MaterialInventory
+ findInventoryByMaterial : MaterialInventory
SupplierAdminRole
+ createWorkArea(userProcessContainer : JPanel, account : UserAccount, organization : Organization, enterprise : Enterprise, business : EcoSystem) : JPanel
+ toString() : String
MaterialShippingRole
+ createWorkArea(userProcessContainer : JPanel, account : UserAccount, organization : Organization, enterprise : Enterprise, business : EcoSystem) : JPanel
+ toString() : String
MaterialInventory
- material : Material
- availableQty : int
- pickingQty : int
+ MaterialInventory(Material)
+ getMaterial
+ setMaterial(Material)
+ getAvailableQty
+ setAvailableQty
+ getPickingQty
+ setPickingQty
+ toString : String
PickingOrderDirectory
- pickingOrderList : ArrayList<PickingOrder>
+ PickingOrderDirectory
+ generatePickingOrderId
+ getPickingOrderList : ArrayList<PickingOrder>
+ setPickingOrderList
+ addPickingOrder : PickingOrder
PickingOrder
- pickingOrderId : String
- materialRequest : MaterialRequest
- organization : SupplierOrganization
+ PickingOrder(String, MaterialRequest, SupplierOrganization)
+ getPickingOrderId
+ setPickingOrderId
+ getMaterialRequest : MaterialRequest
+ setMaterialRequest (MaterialRequest)
+ setOrganization(SupplierOrganization)
+ toString
MaterialRequest
- material : Material
- qty : int
+ MaterialRequest(Material, int)
+ getMaterial : Material
+ setMaterial(Material)
+ getQty
+ setQty (int)
+ toString : String
Material
- materialId : String
- materialName : String
- materialPrice : int
- materialWeight : int
+ Material(String, String, int)
+ getMaterialId
+ setMaterialId
+ getMaterialName
+ setMaterialName
+ toString : String
MaterialCatalog
- materialList : ArrayList<Material>
+ MaterialCatalog
+ getMaterialList : ArrayList<Material>
+ setMaterialList (ArrayList<Material>)
+ addMaterial : Material
ProductionOrganization
- productionPlanDirectory ProductionPlanDirectory
- rawMaterialInventoryDirectory MaterialInventoryDirectory
+ ProductionOrganization()
+ getProductionPlanDirectory ProductionPlanDirectory
+ setProductionPlanDirectory(ProductionPlanDirectory)
+ setRawMaterialInventoryDirectory(MaterialInventoryDirectory)
+ getSupportedRole() : ArrayList<Role>
ProductionAdminRole
+ createWorkArea(userProcessContainer : JPanel, account : UserAccount, organization : Organization, enterprise : Enterprise, business : EcoSystem) : JPanel
ProductionRole
+ createWorkArea(userProcessContainer : JPanel, account : UserAccount, organization : Organization, enterprise : Enterprise, business : EcoSystem) : JPanel
FinishedGoods
- productName : String
- qty : int
- planId : String
+ FinishedGoods (String, int)
+ getPlanId :String
+ setPlanId(String)
+ getProductName : String
+ setProductName (String)
+ getQty : int
+ setQty (int)
+ toString : String
ManufacturerReplenishmentRequest
- medicine : Medicine
- quantity : int
+ ManufacturerReplenishmentRequest(medicine : Medicine, quantity : int)
+ getMedicine() : Medicine
+ getQuantity() : int
+ toString() : String
ProductionOrder
- productName : String
- medicine : Medicine
- qty : int
- planId : String
+ ProductionOrder (String, int)
+ getPlanId :String
+ setPlanId(String)
+ getProductName : String
+ setProductName (String)
+ getQty : int
+ setQty (int)
+ getMedicine() : Medicine
+ setMedicine(medicine : Medicine)
+ toString : String
ProductionPlanDirectory
- productionPlanList : ArrayList<ProductionPlan>
+ ProductionPlanDirectory()
+ generatePlanId() : String
+ getProductionPlanList() : ArrayList<ProductionPlan>
+ setProductionPlanList(productionPlanList : ArrayList<ProductionPlan>)
+ addProductionPlan(productName : String,qty : int) : ProductionPlan
ProductionPlan
- planId : String
- productName : String
- qty : int
- planDate : Date
- status : String
+ ProductionPlan(planId : String,productName : String,qty : int)
+ getPlanId() : String
+ setPlanId(planId : String) : void
+ getProductName() : String
+ setProductName(productName : String)
+ getQty() : int
+ setQty(qty : int) : void
+ getPlanDate() : Date
+ setPlanDate(planDate : Date) : void
+ getStatus() : String
+ setStatus(status : String) : void
+ toString() : String
DeliveryWorkRequest
+ {static} STATUS_RECEIVED..STATUS_REPORTED : String
- pickupLocation : String - dropLocation : String
- cargoDescription : String
- cargoWeightKg : double
- scheduledDate : Date
- assignedTruck : Truck
- assignedDriver : UserAccount
- inspectionResult : String
- proofReference : String
- deliveryReport : String
+ Getters/Setters for all fields
UserAccountDirectory
- userAccountList : ArrayList<UserAccount>
+ UserAccountDirectory()
+ getUserAccountList() :ArrayList<UserAccount>
+ authenticateUser(String username, String password) : UserAccount
+ createUserAccount(String username, String password, Employee employee, Role role) : UserAccount
+checkIfUsernameIsUnique(String username) : boolean
EmployeeDirectory
- employeeList : ArrayList<Employee>
+ EmployeeDirectory()
+ getEmployeeList() : ArrayList<Employee>
+ createEmployee(name : String) : Employee
UserAccount
- username : String
- password : String
- employee : Employee
- role : Role
- workQueue : WorkQueue
+ getUsername() : String
+ setUsername(username : String) : void
+ getPassword() : String
+ setPassword(password : String) : void
+ getEmployee() : Employee
+ setEmployee(employee : Employee) : void
+ getRole() : Role
+ setRole(role : Role) : void
+ getWorkQueue() : WorkQueue
Employee
- name : String
- id : int
- count : int = 1
+ Employee()
+ getId() : int
+ setName(name : String) : void
+ getName() : String
WorkQueue
- workRequestList : ArrayList<WorkRequest>
+ WorkQueue()
+ getWorkRequestList : ArrayList<WorkRequest>
WorkRequest
- message : String
- sender : UserAccount
- receiver : UserAccount
- status : String
- requestDate : Date
- resolveDate : Date
+ WorkRequest()
+ getMessage : String
+ setMessage(String)
+ getReceiver : UserAccount
+ setReceiver : UserAccount
Role
<<enumeration>> RoleType
SupplierAdmin
MaterialShippingOperator
ProductionAdministrator
ProductionOperator
QualityAssurance
TransportAdministrator
Driver
InventoryAdministrator
ShippingOrderStaff
ShippingOperator
ProcurementAdministrator
PharmacyOperator
- value : String
+ getValue() : String

+ createWorkArea(userProcessContainer : JPanel
+ toString() : String
QualityAssuranceOrganization
+ QualityAssuranceOrganization
+ getSupportedRole() : ArrayList<Role>
QualityAssuranceRole
+ createWorkArea(userProcessContainer : JPanel, account : UserAccount, organization : Organization, enterprise : Enterprise, business : EcoSystem) : JPanel
TransportOrganization
- fleetDirectory : FleetDirectory
+ TransportOrganization()
+ getFleetDirectory() : FleetDirectory
+ getSupportedRole() : ArrayList<Role>
TransportAdministratorRole
+ createWorkArea(...) : JPanel
DriverRole
+ createWorkArea(...) : JPanel
FleetDirectory
- truckList : ArrayList<Truck>
+ FleetDirectory()
+ getTruckList() : ArrayList<Truck>
+ addTruck(licensePlate:String, model:String, capacityKg:double) : Truck
+ removeTruck(truck:Truck) : boolean
Truck
+ {static} STATUS_AVAILABLE : String = "Available"
+ {static} STATUS_ASSIGNED : String = "Assigned"
+ {static} STATUS_MAINTENANCE : String = "Maintenance"
- licensePlate : String
- model : String
- capacityKg : double
- status : String
+ Truck(licensePlate:String, model:String, capacityKg:double)
+ getters/setters for all fields
+ toString() : String
InventoryOrganization
- medicineInventoryDirectory : MedicineInventoryDirectory
+ InventoryOrganization()
+ getMedicineInventoryDirectory() : MedicineInventoryDirectory
+ setMedicineInventoryDirectory(medicineInventoryDirectory : MedicineInventoryDirectory) : void
+ getSupportedRole() : ArrayList<Role>
InventoryAdminRole
+ createWorkArea(userProcessContainer : JPanel, account : UserAccount, organization : Organization, enterprise : Enterprise, business : EcoSystem) : JPanel
+ toString() : String
ShippingOrganization
+ ShippingOrganization()
+ getSupportedRole() : ArrayList<Role>
ShippingOperatorRole
+ createWorkArea(userProcessContainer : JPanel, account : UserAccount, organization : Organization, enterprise : Enterprise, business : EcoSystem) : JPanel
+ toString() : String
ShippingOrderStaffRole
+ createWorkArea(userProcessContainer : JPanel, account : UserAccount, organization : Organization, enterprise : Enterprise, business : EcoSystem) : JPanel
+ toString() : String
WholesaleWorkRequest
- requestType : RequestType
- itemDescription : String
- quantity : int
+ WholesaleWorkRequest(requestType : RequestType, itemDescription : String, quantity : int)
+ getRequestType() : RequestType
+ getItemDescription() : String
+ getQuantity() : int
+ toString() : String
ProcurementOrganization
+ ProcurementOrganization
+ getSupportedRole():ArrayList<Role>
ProcurementRole
+ createWorkArea(userProcessContainer : JPanel, account : UserAccount, organization : Organization, enterprise : Enterprise, business : EcoSystem) : JPanel
+ toString() : String
PharmacyOrganization
- medicineInventoryDirectory : MedicineInventoryDirectory
+ PharmacyOrganization()
+ getMedicineInventoryDirectory : MedicineInventoryDirectory
+ setMedicineInventoryList(MedicineInventoryDirectory)
+ getSupportedRole():ArrayList<Role>
PharmacyRole
+ createWorkArea(userProcessContainer : JPanel, account : UserAccount, organization : Organization, enterprise : Enterprise, business : EcoSystem) : JPanel
+ toString() : String
MedicineInventoryDirectory
- inventoryList : ArrayList<MedicineInventory>
+ MedicineInventoryDirectory()
+ getInventoryList : ArrayList<MedicineInventory>
+ setInventoryList(ArrayList<MedicineInventory)
+ addInventory(Medicine, int, int)
+ findInventoryByMedicine(Medicine) :MedicineInventory
MedicineInventory
- medicine : Medicine
- quantity : int
- standardStock : int
+ getMedicine() : Medicine
+ setMedicine (Medicine)
+ getQuantity : int
+ setQuantity(int)
+ setStandardStock(int)
+ getShortageQty : int
+ getSurplus : int
+ isShortage : boolean
+ isExcess : boolean
+ toString : String
MedicineCatalog
- medicineList : ArrayList<Medicine>
+ MedicineCatalog()
+ getMedicineList : ArrayList<Medicine>
+ setMedicineList(ArrayList<Medicine)
+ addMedicine(String, String, int) : Medicine
Medicine
- medicineId : String
- medicineName : String
- medicinePrice : int
+ Medicine(String, String , int)
+ getMedicineId : String
+ setMedicineId(String)
+ getMedicineName : String
+ setMedicineName(String)
+ getMedicinePrice : int
+ setMedicinePrice(int)
+ toString : String
ProcurementRequest
- count : int
- procurementId : int
- medicine : Medicine
- requestQty : int
- replenishmentRequest : ReplenishmentRequest
+ ProcurementRequest()
+ getMedicine : Medicine
+ setMedicine(Medicine)
+ getRequestQty : int
+ setRequestQty (int)
+ getProcurementId : int
+ setProcurementId (int)
+ getReplenishmentRequest : ReplenishmentRequest
+ setReplenishmentRequest(ReplenishmentRequest)
+ getItemDescription: String
+ getQuantity : int
+ toString : String
ReplenishmentRequest
- count : int
- replenishId : int
- medicine : Medicine
- replenishQty : int
+ ReplenishmentRequest()
+ getMedicine : Medicine
+ setMedicine(Medicine)
+ getReplenishQty : int
+ setReplenishQty (int)
+ toString : String